MCA FULL STACK DEVELOPMENT

Practical Report — Question 3

HTML Lists: Ordered, Unordered, Description and Nested Lists

Course	MCA — First Year
Practical	MCA Lab 1, Question 3
HTML file	lists.html
CSS file	style.css

DRAFT — pending browser verification

1. Objective

To create a webpage demonstrating unordered, ordered, description and nested HTML lists, including the ordered-list attributes start and type, with a separate CSS stylesheet.

2. HTML Elements Explained

<ul> creates a bulleted list. **<ol>** creates a numbered or ordered list. **<li>** represents an item in either list. **<dl>** contains a description list, **<dt>** identifies a term, and **<dd>** gives its description. A nested list is a list placed inside a list item of another list.

<header>, **<nav>**, **<main>**, **<section>** and **<footer>** group the webpage into meaningful regions. **<a>** creates links; **href="#nested"** points to an element with a matching ID. **start="5"** starts numbering at five and **type="I"** selects uppercase Roman numerals.

3. CSS Concepts Explained

CSS controls the appearance of the page. Element selectors such as body style the whole page; class selectors such as .content style reusable sections. margin controls outer spacing, padding controls inner spacing, max-width limits content width, border creates an outline, and border-radius rounds corners. The :hover pseudo-class styles links under the pointer. The @media rule adapts spacing for smaller screens.

4. Proposed Project Structure

```
MCA_Full_Stack_Practicals/
■■■ 01_MCA_Lab_1/
    ■■■ Q3_Lists/
        ■■■ lists.html
        ■■■ style.css
        ■■■ README.TXT (recommended)
■■■ 04_Submission_PDFs/
    ■■■ Q3_Lists_Report.pdf
■■■ 05_Output_Screenshots/
    ■■■ Q3_Output_1.png
    ■■■ Q3_Output_2.png
    ■■■ Q3_Output_3.png
```

5. Source Code — lists.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>HTML Lists Practical</title>
    <link rel="stylesheet" href="style.css">
</head>
<body id="top">
    <header class="header">
        <h1>HTML Lists</h1>
        <p>Learning Ordered, Unordered and Description Lists</p>
    </header>

    <nav class="navigation">
        <a href="#unordered">Unordered Lists</a>
        <a href="#ordered">Ordered Lists</a>
        <a href="#description">Description Lists</a>
        <a href="#nested">Nested Lists</a>
    </nav>

    <main>
        <section id="unordered" class="content">
            <h2>1. Unordered List</h2>
            <p>An unordered list displays items with bullets.</p>
            <h3>Programming Languages</h3>
            <ul>
                <li>C</li>
                <li>C++</li>
                <li>Java</li>
                <li>Python</li>
            </ul>
        </section>

        <section id="ordered" class="content">
            <h2>2. Ordered List</h2>
            <p>An ordered list displays items in a sequence.</p>
            <h3>Steps to Create a Webpage</h3>
            <ol>
                <li>Open Visual Studio Code.</li>
                <li>Create an HTML file.</li>
                <li>Write the HTML structure.</li>
                <li>Open the webpage in a browser.</li>
            </ol>
        </section>

        <section id="description" class="content">
            <h2>3. Description List</h2>
            <p>A description list displays terms and their explanations.</p>
            <dl>
                <dt>HTML</dt>
                <dd>Defines the structure of a webpage.</dd>
                <dt>CSS</dt>
                <dd>Controls the appearance and layout of a webpage.</dd>
                <dt>JavaScript</dt>
                <dd>Adds interactivity to webpages.</dd>
            </dl>
        </section>

        <section id="nested" class="content">
            <h2>4. Nested List</h2>
            <p>A nested list contains one list inside another list item.</p>
            <h3>Web Development</h3>
            <ul>
                <li>Frontend Development
                    <ul>
                        <li>HTML</li>
                        <li>CSS</li>
                        <li>JavaScript</li>
                    </ul>
                </li>
                <li>Backend Development
                    <ul>
                        <li>Node.js</li>
                        <li>Python</li>
                        <li>PHP</li>
                    </ul>
                </li>
            </ul>
        </section>

        <section class="content">
            <h2>5. Ordered List Attributes</h2>
            <h3>List Starting from 5</h3>
            <ol start="5">
                <li>Fifth item</li>
                <li>Sixth item</li>
                <li>Seventh item</li>
            </ol>
            <h3>Uppercase Roman Numerals</h3>
            <ol type="I">
                <li>Introduction</li>
                <li>Development</li>
                <li>Conclusion</li>
            </ol>
        </section>

        <section class="content resources">
```

```
            <h2>Useful Resources</h2>
            <p>Learn more about HTML lists on
                <a href="https://developer.mozilla.org/en-US/docs/Web/HTML/Element/ul"
                   target="_blank" rel="noopener noreferrer">MDN Web Docs</a>.
            </p>
            <p><a href="#top">Back to the top</a></p>
        </section>
    </main>

    <footer class="footer">
        <p>HTML Practical — Question 3</p>
    </footer>
</body>
</html>
```

6. Source Code — style.css

```
body {
    font-family: Arial, sans-serif;
    background-color: #f1f5f9;
    color: #1e293b;
    margin: 0;
    line-height: 1.6;
}

.header {
    background-color: #1e3a5f;
    color: white;
    text-align: center;
    padding: 30px 15px;
}

.navigation {
    background-color: #0f172a;
    padding: 15px;
    text-align: center;
}

.navigation a {
    color: white;
    text-decoration: none;
    margin: 0 12px;
    font-weight: bold;
}

.navigation a:hover {
    text-decoration: underline;
}

.content {
    max-width: 900px;
    margin: 25px auto;
    padding: 22px;
    background-color: white;
    border: 1px solid #cbd5e1;
    border-radius: 8px;
}

.content h2 {
    color: #1e3a5f;
}

ul, ol {
    padding-left: 30px;
}

li {
    margin-bottom: 8px;
}

dt {
    font-weight: bold;
    color: #075985;
    margin-top: 12px;
}

dd {
    margin-left: 25px;
    margin-bottom: 12px;
}

.resources a {
    color: #075985;
}

.footer {
    background-color: #1e3a5f;
    color: white;
    text-align: center;
    padding: 15px;
}

@media (max-width: 600px) {
    .content {
        margin: 15px 10px;
        padding: 15px;
    }

    .navigation a {
        display: inline-block;
        margin: 5px;
    }
}
```

7. Browser Verification Checklist

- Open lists.html in a browser.
- Confirm CSS is loaded.
- Unordered list displays bullets.
- Ordered list displays numbers.
- Description list displays terms and definitions.
- Nested lists appear under their parent items.
- start="5" and type="I" work correctly.
- Internal navigation links jump to the right sections.
- External MDN link opens as expected.
- Layout is readable on a narrow screen.

8. Result

The provided source code is intended to demonstrate the listed HTML list types and basic CSS styling. This report remains a draft until the student runs the page, verifies the behavior and adds genuine browser screenshots. Do not submit it as evidence of successful testing before completing those steps.

9. GitHub Submission

```
git status
git add .
git commit -m "Add Q3 HTML lists practical"
git push
```